

Symmetry Method Homework 6

1). For $SU(2)$, use the fact that

$$\hat{L}_3 |j, m\rangle = m |j, m\rangle \quad \& \quad \hat{L}_{\pm} |j, m\rangle = C_{j, m}^{\pm} |j, m \pm 1\rangle$$

to write down the matrices for $j=1$ representation:

$$\hat{L}_3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix}, \quad \hat{L}_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, \quad \hat{L}_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 & -i & 0 \\ i & 0 & -i \\ 0 & i & 0 \end{pmatrix}$$

2). For $SU(2)$, use the fact that

$$\hat{L}_3 |j, m\rangle = m |j, m\rangle \quad \& \quad \hat{L}_{\pm} |j, m\rangle = C_{j, m}^{\pm} |j, m \pm 1\rangle$$

to write down the matrices for $j=3/2$ representation:

3). For the Simple Harmonic Oscillator, the algebra is

$$[\hat{H}, \hat{a}] = -\hbar\omega \hat{a}, \quad [\hat{H}, \hat{a}^{\dagger}] = \hbar\omega \hat{a}^{\dagger}, \quad [\hat{a}, \hat{a}^{\dagger}] = 1$$

Write down the weight diagram for the representation.

4). For $SU(2)$, what is the roots of $\hat{L}_+$ & $\hat{L}_-$?

5). For $SU(2)$, write down the weight diagram for

5.1) $j=3/2$ representation

5.2) $j=2$ representation

6). Show that for Pauli matrices, $\hat{\sigma}_i$,

$$6.1) \quad [\hat{\sigma}_i, \hat{\sigma}_j] = 2i \sum_k \epsilon_{ijk} \hat{\sigma}_k$$

$$6.2) \quad \{\hat{\sigma}_i, \hat{\sigma}_j\} \equiv \hat{\sigma}_i \hat{\sigma}_j + \hat{\sigma}_j \hat{\sigma}_i = 2\delta_{ij} \hat{1}$$

$$6.3) \quad \text{Tr}(\hat{\sigma}_i \hat{\sigma}_j) = 2\delta_{ij}$$